JEFF MUTUKU

Jffmutuku@gmail.com | linkedin.com/in/jeffmutuku | github.com/JuvoLab
Performance Scientist | Exercise Physiology | Sports Science & Research Analytics

PROFESSIONAL SUMMARY

Performance scientist combining physiological assessment, NCAA Division I performance support, collaborative human-subject research, and analytical reporting. Experience includes metabolic and body-composition testing, athlete monitoring, study methodology, data collection and interpretation, and independent dashboard development.

EDUCATION

California Baptist University **May 2024 - August 2026**
Master of Science in Kinesiology - Concentration: Exercise Science | GPA: 3.89 / 4.00

Notre Dame College **Fall 2019 - December 2022**
Bachelor of Arts in Sports Management | GPA: 3.70 / 4.00 | Cum Laude; Dean's List

PROFESSIONAL EXPERIENCE

Performance Scientist **May 2026 - Present**
Wellth Performance

- Conduct resting metabolic rate and maximal exercise assessments; evaluate oxygen consumption, ventilatory responses, and aerobic capacity. Perform DEXA and 3D body-composition assessments and communicate findings.
- Develop AI-assisted metabolic analysis and reporting workflows using predefined physiological criteria to identify candidate ventilatory thresholds and estimate training zones.
- Contribute to retrospective body-composition research, data organization, methodological review, and scientific reporting.

Performance Coach Intern **July 2025 - May 2026**
Athletic Performance Center, California Baptist University

- Supported performance operations across 12 NCAA Division I programs, working with strength and conditioning staff and sport-specific coaches.
- Conducted force-plate countermovement jump assessments and supported workload monitoring using STATSports GPS and accelerometry.

RESEARCH EXPERIENCE

Performance Researcher & Co-Principal Investigator **January - May 2026**
Post-Activation Potentiation & Sprint Swimming - Master's Thesis Support

- Supported collaborative research in 12 Division I swimmers and 72 experimental trials, contributing to data collection, data hygiene, and study work involving takeoff power and 25-yard sprint performance.
- Independently initiated and developed an interactive HTML/CSS/JavaScript and Plotly dashboard for group and individual response comparisons.

Co-Principal Investigator **January - May 2026**
Seasonal Variations of Heart Rate Recovery in Division I NCAA Swimmers

- Contributed to recruitment, methodology, testing, data quality, analysis and interpretation, and conclusions in an 11-swimmer seasonal study using Bruce-protocol testing and exponential recovery modeling.
- Research poster presented at California Baptist University's University Research Colloquium, April 2026.

Co-Principal Investigator **January - May 2025**
Low-Frequency BFR Walking & Cardiometabolic Markers

- Contributed to recruitment, methodology, data collection, data quality, interpretation, and conclusions in a four-week BFR walking study with 20 participants.
- Research poster presented at California Baptist University's University Research Colloquium, April 2025.

TECHNICAL EXPERTISE

Research & testing: Human-subject protocols, participant screening, informed consent, testing standardization, DEXA, indirect calorimetry, force plates, GPS and accelerometry.

Analysis & development: Python (Pandas, NumPy, SciPy), exponential modeling, data quality, Plotly, HTML/CSS, JavaScript, Excel, Power BI, Streamlit, Git/GitHub and Vercel.

CREDENTIALS & PROFESSIONAL DEVELOPMENT

NASM-CPT | California Department of Public Health X-Ray Technician Permit | STATSports Certified GPS User | FIFA Diploma in Football Medicine | NSCA Member | CPSS Exam Candidate (2026; certification in progress)